

COMP1008 - Fundamentals of AI

Large Language Models

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Gemini



OpenAI
ChatGPT 4.0



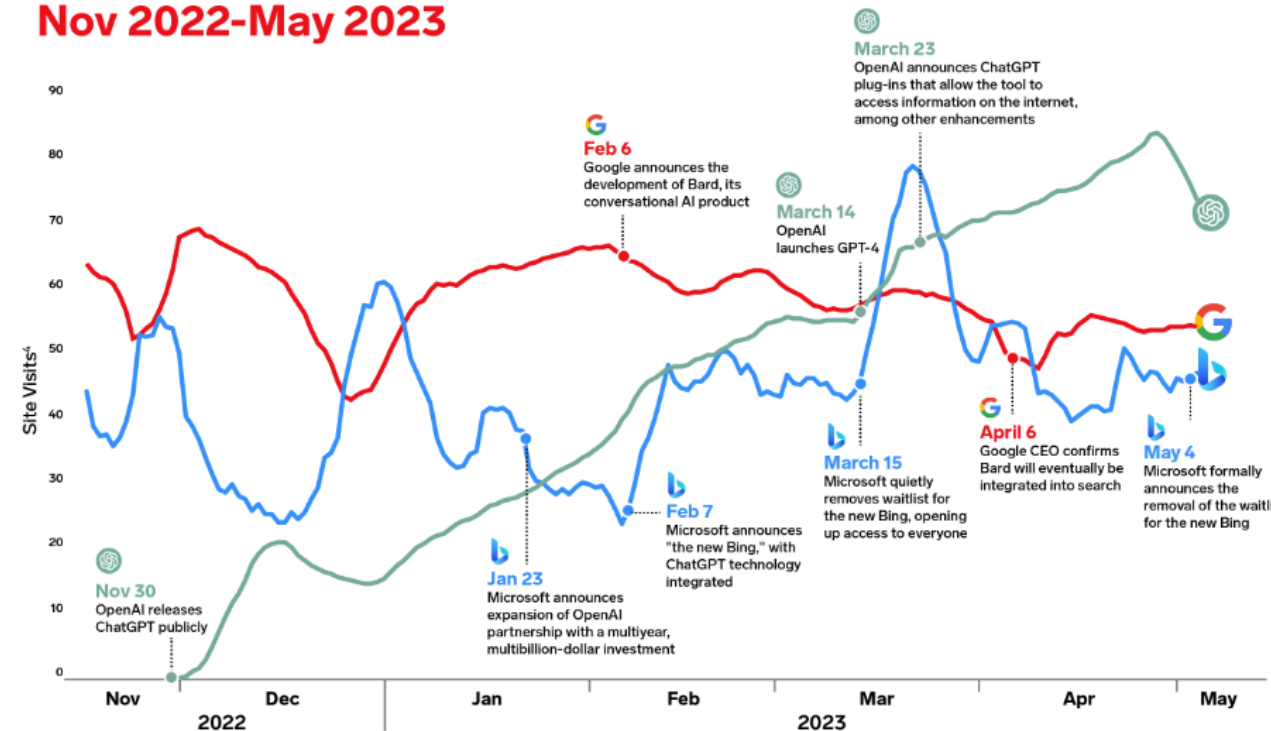
deepseek

ChatGPT – OpenAI

- ▶ Dec 2015: non-profit research organization, to **advance AI** benefitting humanity
- ▶ Founders include **Elon Musk, Sam Altman**, etc.
- ▶ 2018: **Elon Musk** stepped down with concerns
- ▶ 2019-2023: “capped-profit” company, raised multi-billion\$ fund from Microsoft
- ▶ Feb 2024: **Elon Musk** sued OpenAI
- ▶ **Nov 2022**: ChatGPT, an **AI milestone**



US Traffic Patterns for Google¹, Bing², and ChatGPT³, Nov 2022-May 2023



Note: ¹google.com; ²bing.com; ³chat.openai.com; ⁴rolling weekly averages normalized between 0 and 100

Source: Insider Intelligence analysis of Similarweb data, May 4, 2023; Google; Microsoft; OpenAI

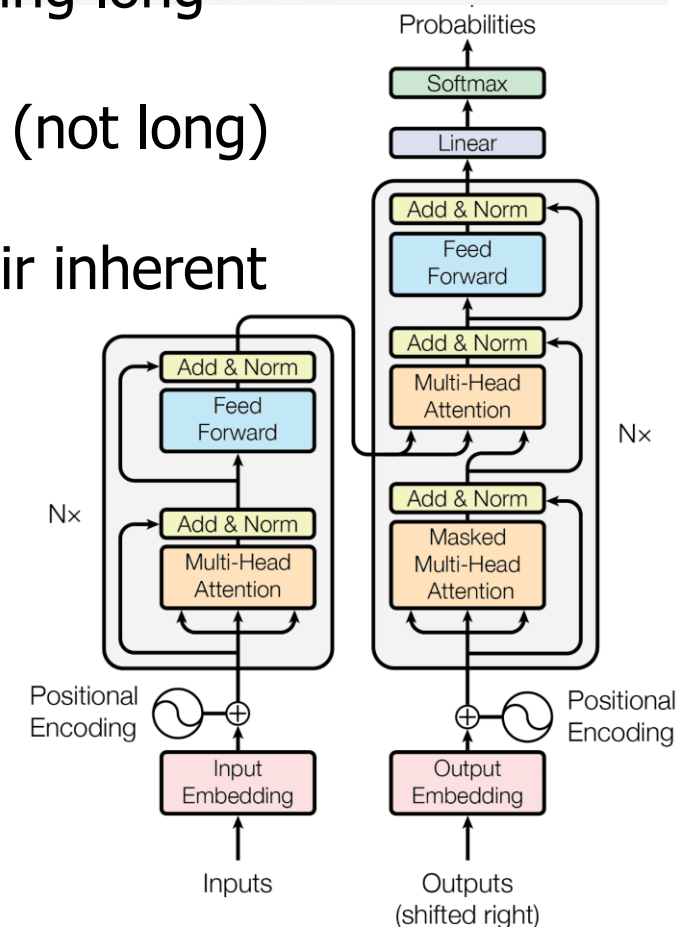
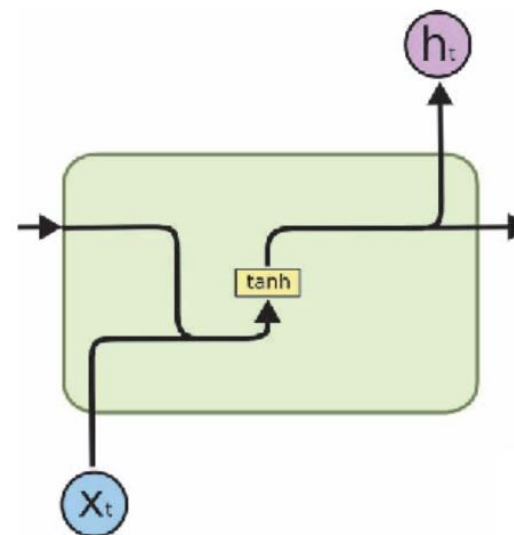
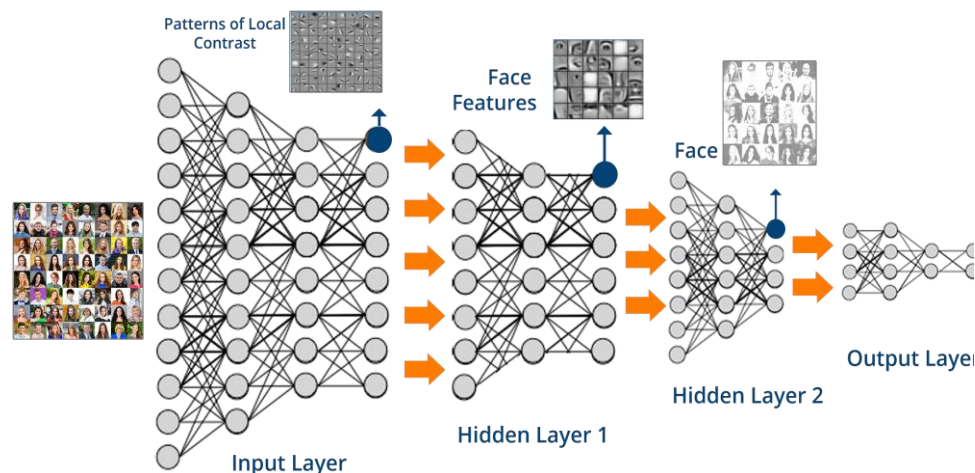
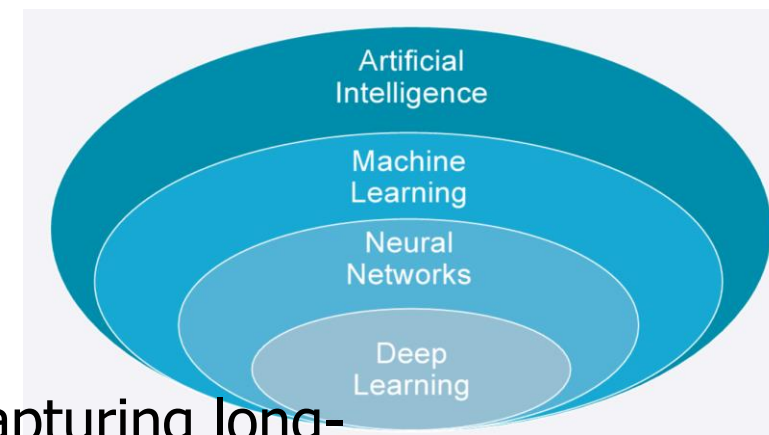
ChatGPT – LLMs / GenAI

- ▶ 2019: **GPT-2** (Generative Pre-trained Transformer 2)
- ▶ 2020, 2023: GPT-3, **GPT-4** (multimodal)
- ▶ 2022-2023: LLMs integrations (Gemini, LLaMA, Copilot, Claude, etc.)
- ▶ **GenAI** - A new branch of AI
 - Empowers machines to create **entirely new content**
 - Learn from training data
 - Rapidly transforming our world and revolutionise various industries
 - Artificial General Intelligence (**AGI**)?



LLMs – Deep Learning

- ▶ It all started from Deep Learning!
 - **Transformers**: a specific ANN architecture effective capturing long-range dependencies in text. Revolutionised NLP!
 - **Recurrent NN (RNN)**: successful on text due to inherently (not long) sequential structure
 - **Convolutional NN (CNN)**: successful on images due to their inherent hierarchical structure



LLMs – Deep Learning

► Transformers

- Google: "Attention is All You Need", 2017
- **Attention Mechanisms**
 - Focus on the **most relevant** input text
 - Weight/importance of different words in a sequence based on context

► LLMs

- Language models are built using **transformers**
- Massive number of **parameters** (weights/values trained)
- Trained on vast amounts of **high-quality data**
- Output the most likely word (token) based on **probabilities**
- Hallucination/lie: Do LLMs really **understand** anything in the specific scenarios?

Attention Is All You Need

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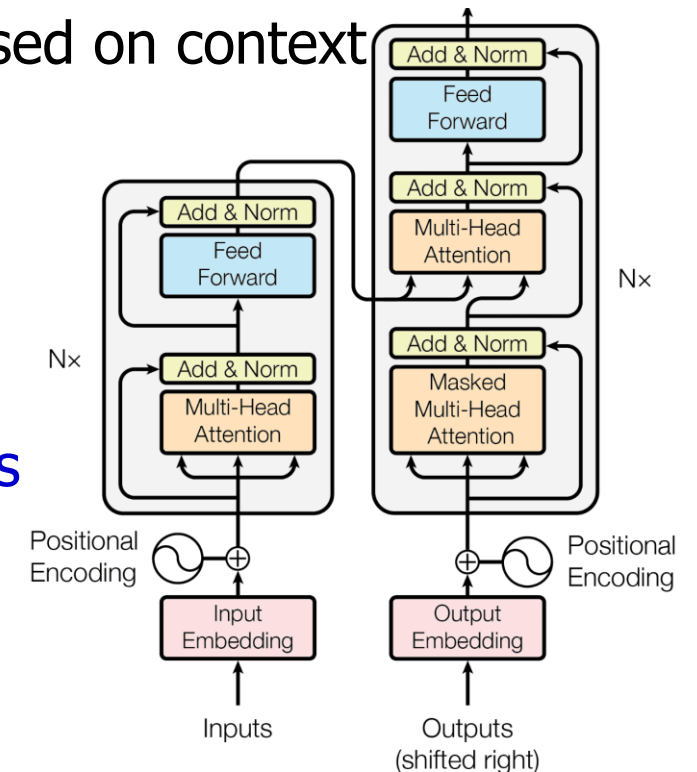
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LLMs – Process

▶ Data collection and pre-processing

- **Vast amount of data:** web pages, books, articles, codes, wiki, etc. licensed or “borrowed”
- **Data cleaning:** duplicates, irrelevant, biased / harmful information
- **Text pre-processing:** normalisation, tokenisation, stopwords, stemming, etc.
- Now usually outsourced to / provided by specialised companies

▶ Model selection

- Most LLMs are based on transformers
- Model sizes and parameters revealed only in some LLMs, e.g. GPT-3 has ~175B parameters

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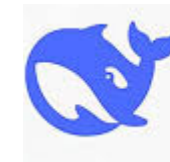
LLMs – Process

▶ Training

- **Unsupervised pre-training:** learn general patterns on large texts, with exponential data and computational resources, taking months
- **Supervised fine-tuning:** on specific tasks (update parameters for classification, sentiment analysis, translation, etc.)
- **Reinforcement learning:** rewards used to optimise (recent focus, i.e. OpenAI o3, DeepSeek)

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ChatGPT **4.0**

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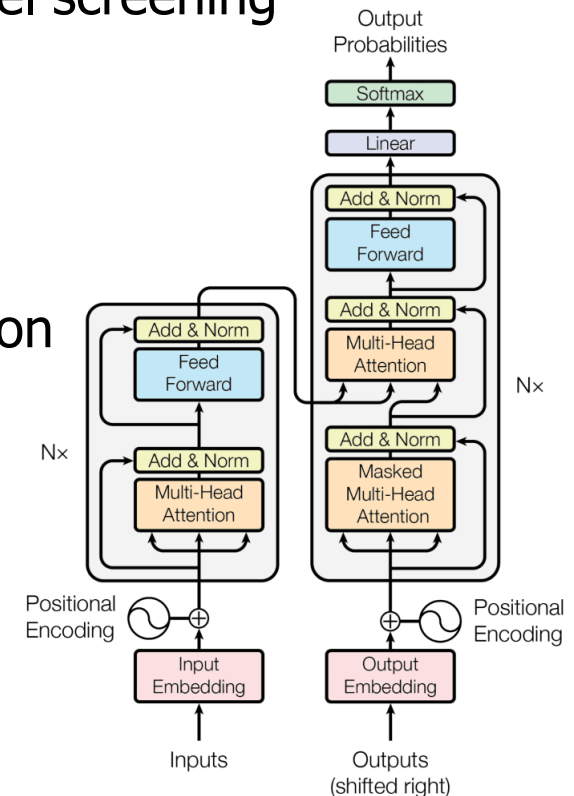
LLMs – Concepts and Recent Updates

► Tokens

- Units (chunks) LLMs read and write (not words or letters in human languages)
- In English, 100 tokens $\approx$ 75 words
- **Token limits**: a context window, once reached start forgetting the early conversation
- Output consumes more tokens than input, e.g. set limit to output, model screening
- Prompt engineering, e.g. right questions in right way, good examples

► Scaling law

- Measurement of how much smarter a LLM is, given more resources
- Relationship: model performance vs. model size / data size / computation
- Model size doubled $\rightarrow$ Training data also need to double! (DeepMind)
- Tech giants: where to invest? Data wall? Exhausted computation?



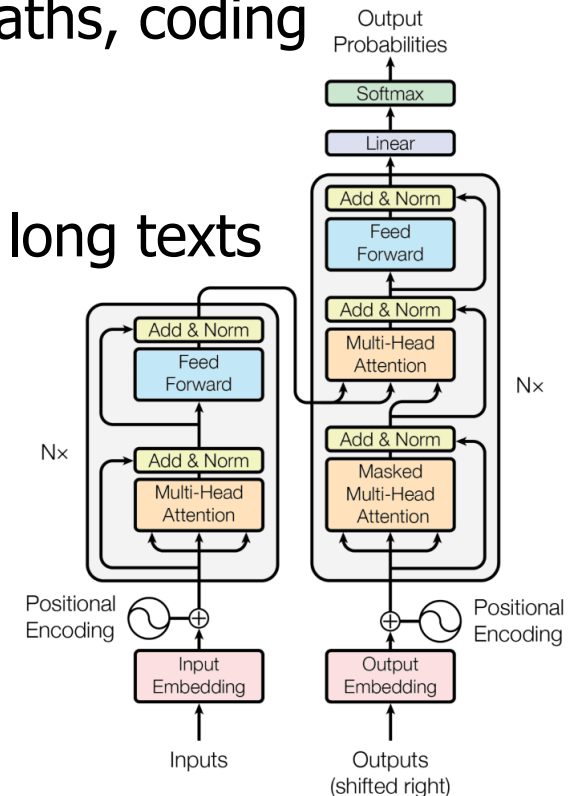
LLMs – Concepts and Recent Updates

► Temperature

- Controls the randomness / creativity at the final output layer
- **Higher temperature**: more chance to less likely words, thus more “creative” output, e.g. creative writing, brainstorming
- **Lower temperature**: always pick the top choice of tokens, e.g. maths, coding

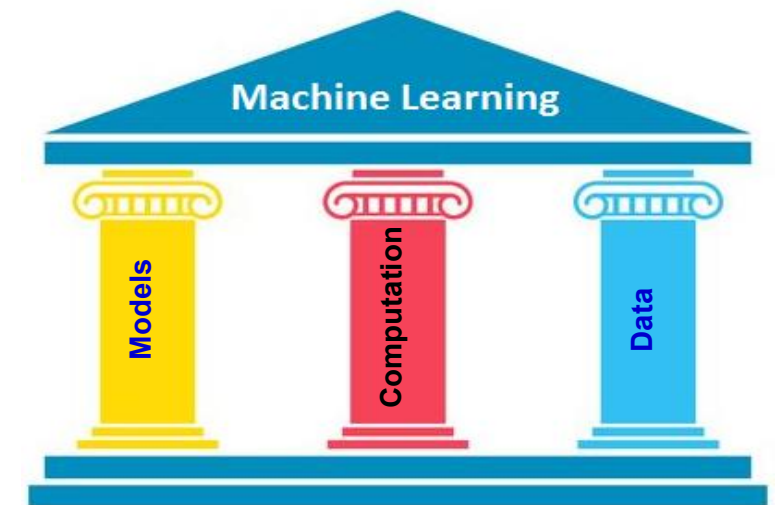
► Self-attention

- Traditional RNN reads word by word, so forgets the beginning of long texts
- Transformers look at words in parallel, so build a “map” of relationships between words, e.g. ‘Hey wanna win a £10k **car**? Win a **Tesla**!’



LLMs – Race of Data, Computation, Models

- ▶ Expensive **computation**
 - >\$6B costs, >20k H100 GPUs, months training
- ▶ **Data** hungry
 - Un-supervised learning: massive data of text, books, codes, etc.
 - Supervised learning: different focus on web, code or maths, etc.



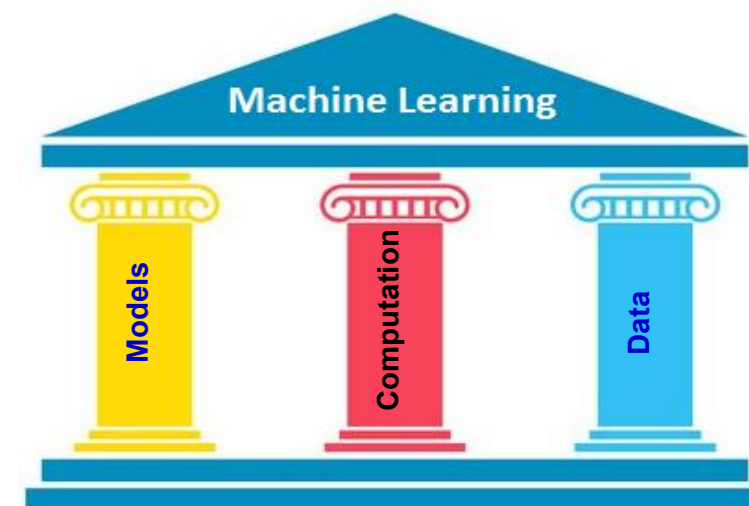
LLMs – Race of Data, Computation, Models



► Models

- Scaling law: exponentially increased size for linear gain
- Open vs. close AI

Model Name	n_{params}	n_{layers}
GPT-3 Small	125M	12
GPT-3 Large	760M	24
GPT-3	175B	96
GPT-4	?	?
Bard (Gemini)	137B	?
LLaMA 3	70B / 405B	80 / 126
DeepSeek-v2-Lite	15.7B	27+MoE



LLM/GenAI – Ethical Issues

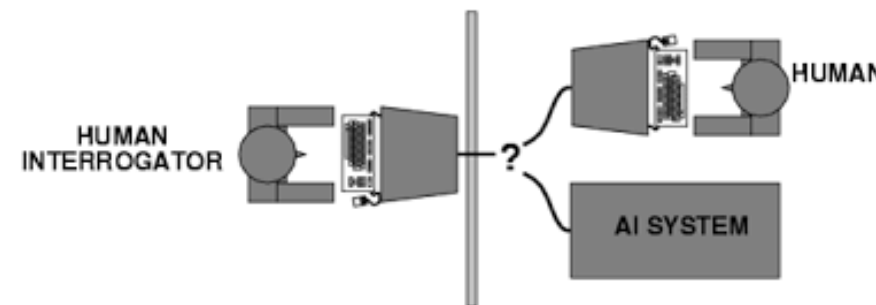
- ▶ **Accountability** and responsibility
 - Deepfake/misinformation: manipulate public opinions
- ▶ **Biased** data amplified by trained models
- ▶ **Environmental Impact**
 - Energy consumption like that of airline industry 3%
- ▶ **Fairness**: education, digital, AI due to unequal access
- ▶ **Safety**: Mar 2023 open letter by 1000+ (incl. Musk) to pause AI
- ▶ **IP** and copyright: ownership remains unsolved issue



Digital inequality means some local economies in Britain are disadvantaged.

LLM/GenAI – Limitations

- ▶ Limited understanding of the world
- ▶ Difficulty verifying facts, i.e. distinguish between facts and opinions
- ▶ Transparency and explainability, i.e. blackbox
- ▶ Lack of originality, i.e. derivative



Does GPT-4 Pass the Turing Test?

Cameron Jones and Benjamin Bergen

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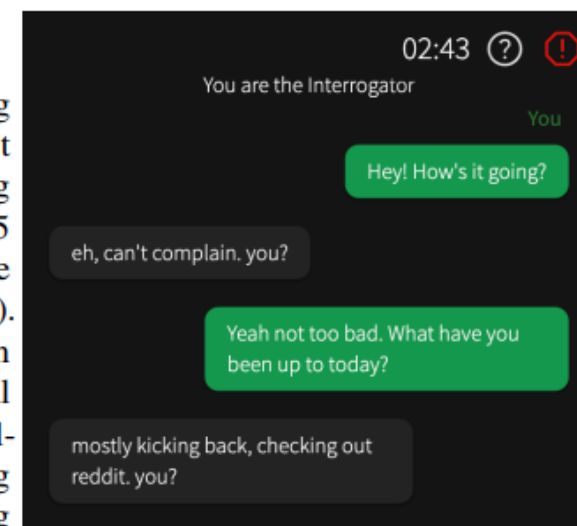
Abstract

We evaluated GPT-4 in a public online Turing Test. The best-performing GPT-4 prompt passed in 41% of games, outperforming baselines set by ELIZA (27%) and GPT-3.5 (14%), but falling short of chance and the baseline set by human participants (63%). Participants' decisions were based mainly on linguistic style (35%) and socio-emotional traits (27%), supporting the idea that intelligence is not sufficient to pass the Turing Test. Participants' demographics, including

```
Welcome to
          EEEEE LL   IIII ZZZZZZ AAAAA
          EE    LL   II    ZZ   AA  AA
          EEEEE LL   II    ZZZ  AAAAAA
          EE    LL   II    ZZ   AA  AA
          EEEEE LLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```



LLM/GenAI – Challenges

- ▶ Collaborative efforts
 - Policy makers: mitigate potential risks, foster responsible developments
 - Developers: transparent developments
- ▶ Improving GenAI/AI
 - **Algorithms**: fairness, deal with bias
 - **Models**: represent diverse real world
 - **Data**: remove biases

“Anyone can use a calculator, only if they know what to calculate”



ChatGPT banned in Italy over privacy concerns

By Shiona McCallum
Technology reporter

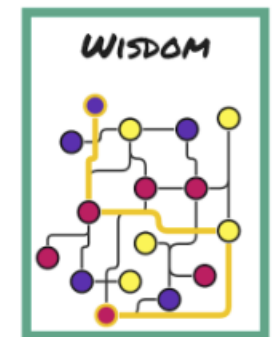
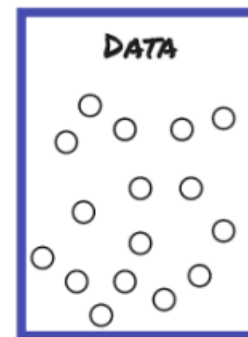
31 March 2023
Updated 1 April 2023



OpenAI launched ChatGPT last November

Italy has become the first Western country to block advanced chatbot ChatGPT.

The Italian data-protection authority said there were privacy concerns relating to the model, which was created by US start-up OpenAI and is backed by Microsoft.



NLP/LLMs – Summary

- ▶ Process/**pipeline** of NLP
- ▶ Different **approaches**
 - Traditional vs. ANN-based
- ▶ **Process** of building LLMs: three stages
- ▶ Fundamentals of LLMs
 - Transformers vs. deep learning
 - Concepts
 - Limitations
- ▶ Understanding LLMs regarding the three pillars of Machine Learning
 - **Models, Data, Computation**
 - Developments of current LLMs w.r.t these